REAL TIME FACE MASK DETECTION USING PYTHON

Developed Using Visual Studio Code

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INTRODUCTION

- Face mask detection is a real-time application that identifies whether a person is wearing a mask or not using computer vision
- Face mask detection systems can be used to monitor whether the people are wearing masks.
- This project is crucial for public safety and health monitoring ,especally in crowded areas
- This model is trained by face with mask and without mask image.

PROBLEM STATEMENT & OBJECTIVES

- Build a real-time face mask detection system.
- Deep learning has gained further attention in object discovery and was used for mortal discovery purpose.
- It develops a face mask discovery tool that can discrete whether the existent is wearing mask or not.
- To reduce the possibility of infection ,it’s advised that people should wear mask and maintain a distance of at least 5feet from each other.
- Build a real-time face mask detection system.
- Implement and test the system in Visual Studio Code.
- Achieve accurate and fast predictions.

MECHANISM OF FACE RECOGNITION

- Image is needed to be captured.
- The position of the eyes are determined.
- The image is converted to grayscale and resized.
- The image is turned into a prototype that the search engine users to produce facial comparison results.
- To compare the template to other templates on disk,a sophisticated algorithm is used to search and match the image.

TOOLS & TECHNOLOGIES

- Programming Language python.
- IDE : Visual Studio Code.
- Libraries :OpenCV , TensorFlow/ Keras, NumPy
- Hardware : Laptop or Webcam

DATASET USED

- Dataset from Kaggle containing images with and without masks.
- Categories : With Mask , Without Mask
- Preprocessing: Resizing, normalization , augmentation
- Balanced and labeled data.

MODEL ARCHITECTURE

- A Convolution Network(CNN) was used:
- --Convolution + Relu Layers
- --Max Pooling layers
- --Fully Connected (Dense) layers
- --Dropout for regularization
- --Soft max for final classification.

System Workflow

1. Open webcam using OpenCV
2. Detect faces using Haar Cascade
3. Crop and preprocess face ROI
4. Pass to CNN model
5. Predict and annotate result on screen.

Code in Visual Studio Code

- Project organized into modules
- Main.py for running webcam stream
- Model.py for CNN architecture
- Utiles.py for processing and helper functions
- Execution and debugging done via VS Code terminal.

RESULTS

- Achieved over 95% accuracy
- Real-Time detection with boundary boxes
- Screenshots show clear label : Mask/No mask.

CHALLENGES FACED

- Detecting masks in low-light conditions
- Handling partial occlusion
- Real-Time performance optimization
- Dataset imbalance between classes.

FUTURE ENHANCEMENTS

- Add detection for improperly worn masks
- Deploy to Raspberry Pi or mobile devices
- Integrate with access control systems
- Use more robust face detection techniques
- This can brought ease to control covid-19 by using it with alarm or by charging from the people who had not wear face mask.

CONCLUSION :-

- Face mask detection using python and deep learning can be a powerful tool for health and safety.
- Visual Studio Code provides a flexible platform for developing and debugging the application.


Thank
you!